Certified whole-domain bounds and equality barriers for the odd theta–Weil form

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Abstract

We prove explicit lower bounds on the full complex odd form domain of a theta-weighted realization of the Riemann–Weil form. After removing any closed verified kernel containing the score, an admissible residual-enriched trial proves a family of explicit bounds. One retains 0.132584532412 of a positive gamma–prime-symbol energy with loss less than 0.034762; another retains 0.212458845389 with loss less than 0.039482. Among the positive-retention certificates in a set of 231 tested parameter pairs, the least scalar loss is 0.031324476155738682727829, with energy coefficient 0.000118321042; no global optimality is asserted. The certificate keeps the entire physical residual Gram, its coupling outside the trial space, all prime powers, the contact and pole terms, and the infinite remainders. The exact obstruction frame has physical rank 56. We also prove that the fixed positive midpoint comparison overshoots an actual verified equality vector by more than 7/5000 in the original Hilbert norm. This gives a quantitative barrier to zero-loss certification by that comparison, even with perfect trial enrichment, and an exact kernel-aware replacement. The operator algebra uses established residual and factorization principles; the contribution is their fully priced theta–arithmetic realization and its equality obstruction. Exact continuum cancellation also gives a complete signed-action error below $11 \cdot 10^{-6}$ for the enriched trial, with every prime tail and pole retained. No zero-error positivity, even-sector conclusion, or resolution of RH is claimed.

1 The result and its scope

Put $\Xi(t) = \xi(1/2 + it)$, with $\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)/2$. For $x \geq 0$ define

$$\Phi(x) = \sum_{n \geq 1} e^{x/2 - \pi n^2 e^{2x}} \left(4\pi^2 n^4 e^{4x} - 6\pi n^2 e^{2x}\right), \qquad \Phi(-x) = \Phi(x).$$

The theta transformation gives $\widehat{\Phi} = \Xi$ for $\widehat{h}(t) = \int_{\mathbb{R}} h(x) e^{itx}\, dx$. Set $w^2 = \Phi/\cosh(x/2)$, $p = w \sinh(x/2)$ and $H = H_{\mathrm{odd}} \subset L^2(\mathbb{R}, dx)$. The equivalent positive-half-line measure is $2\,dx$ throughout. Let $b(t) = \Re\psi(1/4 + it/2) - \psi(1/4)$ and let G be the positive closed form operator $M_w b(D) M_w$. Its form domain is

$$\mathcal{V} = \left\{ u \in H : \frac{1}{2\pi} \int b(t) |\widehat{wu}(t)|^2\, dt < \infty \right\}.$$

The ordinary signed odd form is

$$\mathfrak{a}(u) = g(u) + c_\Gamma \|wu\|^2 - \langle Ju, u\rangle - 2|\langle u, p\rangle|^2, \quad c_\Gamma = \psi(1/4) - \log\pi, \tag{1}$$

where the complete norm-convergent prime-power operator is

$$J = \sum_{n\geq 2} \frac{\Lambda(n)}{\sqrt{n}} M_w (T_{\log n} + T_{-\log n}) M_w.$$

Here $T_a h(x) = h(x+a)$; both shifts make the convention immaterial to the selfadjoint sum. The contact constant, gamma symbol, all prime powers and odd pole in (1) are retained. The even pole disappears only because the present space is odd. The signed realization originates in [1, Lemma 5.1]. Supplement S0 proves the domain, kernel and tail facts additionally used here.

Define $q = \Phi'/w$ and $Q = \|q\|^2$. The score is in $\mathrm{Dom}(G) \cap \ker A_W$, where A_W represents (1). The positive energy used in this paper is

$$\begin{aligned} E_8(u) &= \frac{1}{2\pi} \int d_{8,+}(t) |\widehat{wu}(t)|^2 \, dt, \\ d_8(t) &= m_\Gamma(t) - k_8(t), \\ m_\Gamma(t) &= \Re\psi(1/4 + it/2) - \log\pi, \\ k_8(t) &= 2 \sum_{n=2,3,4,5,7,8} \frac{\Lambda(n)}{\sqrt{n}} \cos(t\log n). \end{aligned} \tag{2}$$

This is a different positive form from the score-difference energy e_0 associated with the score. No ordering between them is assumed.

Theorem 1.1 (Whole-domain bounds)**.** *For every closed verified odd subspace $K \subset \ker A_W$ containing q, every complex $u \in \mathcal{V}$, and $z = (I - P_K)u$, we prove*

$$\begin{aligned} \mathfrak{a}(u) &\geq 0.132584532412 E_8(z) - 0.034762\|z\|^2, \\ \mathfrak{a}(u) &\geq 0.212458845389 E_8(z) - 0.039482\|z\|^2, \\ \mathfrak{a}(u) &\geq \frac{59160521}{500000000000} E_8(z) - \frac{31324476155738682727829}{10^{24}} \|z\|^2. \end{aligned}$$

There are no further finite orthogonality conditions. All constants use the original physical norm and are independent of the equality-frame index m. Their pointwise maximum is valid.

The first two inequalities simultaneously improve the loss and energy coefficient of the respective earlier bounds $0.03E_8 - 0.077614688545235\|\cdot\|^2$ and $0.18E_8 - 0.08208813127141\|\cdot\|^2$. The last bound has the smallest scalar loss among the tested candidates with strictly positive energy retention, but its smaller energy coefficient prevents a scalar-only ordering. A negative lower bound does not establish positivity. For $z \neq 0$, write $r = E_8(z)/\|z\|^2 \geq 0$. Each assertion is an ordinary affine lower bound in r. At small r the least-loss row is useful; at larger r a row retaining more energy is stronger. Taking the maximum is a pointwise operation on these scalar bounds, not the construction of a new quadratic form. The equality case $z = 0$ gives $\mathfrak{a}(u) = 0$ by the verified kernel identity.

All finite matrices in the construction act on complex coefficient spaces. Their stored entries happen to be real; a real symmetric Loewner comparison extends to complex vectors by applying it to their real and imaginary parts. The contour and form arguments below likewise use sesquilinear pairings. Thus the complex quantifier is part of the proof, rather than an inference from tests on real trial vectors.

1.1 Normalization and the meaning of the quotient

The norm in Theorem 1.1 is

$$\|z\|^2 = \int_{\mathbb{R}} |z(x)|^2\,dx = 2\int_0^\infty |z(x)|^2\,dx.$$

If the physical Weil test is written as $f = wz$, this norm is $\int |f(x)|^2/w(x)^2\,dx$. The weight tends rapidly to zero, so it is essential to state which of these two variables carries the comparison norm. Multiplication by w also does not impose compact support. All Fourier cutoffs used in the computation are accompanied by tails in this original weighted realization.

The word "verified" in the kernel hypothesis means that membership in the operator kernel has been established for that subspace. It asserts no completeness of the currently known equality vectors. The smallest permitted choice is $K = \operatorname{span}\{q\}$. If K is enlarged, P_K remains the actual Hilbert orthogonal projection; an approximate frame projection cannot be substituted. The exact signed form annihilates every mixed pairing with K, so

$$\mathfrak{a}(u) = \mathfrak{a}(z), \qquad z = (I - P_K)u.$$

Moreover $z \in \mathcal{V} \cap q^\perp$. Thus one certificate on the score quotient proves the displayed inequality for every such K. This transfer does not assert that the positive energies of two successive quotients are ordered, and it does not transfer a numerical Gram or conditioning bound onto a newly projected frame.

There are three distinct spaces in the finite construction. The coefficient space of B has dimension 2906; the exact map $C = BZ$ has 56 input coordinates; and its vectors lie in the infinite physical Hilbert space. The coefficient metric $N = Z^*Z$ measures $\|Zx\|^2$, whereas $H = C^*C$ measures $\|Cx\|^2$. The proof retains both. This distinction is particularly important because the physical frame has a much smaller lower Gram bound than its coefficient frame. Numerical orthogonality in one metric cannot supply an inverse or a form comparison in the other.

2 A complete matrix residual certificate

The next lemma records the precise reusable certificate used below. It is an operator-valued version of standard coercive residual certification; see [3, Section 4.3, Proposition 4C].

Lemma 2.1. *Let $E \geq 0$ be selfadjoint on a Hilbert space, $s > 0$, and C a finite column map. Let $N > 0$ be a coefficient matrix and $\nu > 0$. For a trial map Y_0 with columns in* $\operatorname{Dom}(E)$*, suppose a selfadjoint proxy E_m is defined on its range and $\|(E - E_m)|_{\operatorname{ran} Y_0}\| \leq \eta$. Set*

$$Q_0 = Y_0^*Y_0, \quad r = C - (E_m + s)Y_0, \quad S_0 = r^*r, \quad K_m = C^*Y_0 + Y_0^*C - Y_0^*(E_m + s)Y_0.$$

For any $\varepsilon > 0$, the matrix comparison

$$\nu \left[K_m + \left(\eta + \frac{(1+1/\varepsilon)\eta^2}{s} \right) Q_0 + \frac{1+\varepsilon}{s} S_0 \right] \le \theta N \tag{3}$$

implies $\nu C N^{-1} C^* \le \theta(E+s)$ as forms. It retains the entire residual in the ambient Hilbert space. If symmetric centers K_c, Q_c, S_c have full operator errors k_e, q_e, s_e, adding $\nu[k_e + (\eta + (1+1/\varepsilon)\eta^2/s)q_e + (1+\varepsilon)s_e/s]I$ to the center matrix suffices.

Proof. Put $T = E + sI$. Selfadjointness and $T \ge sI$ give a bounded inverse with $0 \le T^{-1} \le s^{-1}I$. This is precisely where the strictly positive shift is used. Since each trial column belongs to $\mathrm{Dom}(T)$, the maps TY_0 and $R = C - TY_0$ are well defined in the ambient Hilbert space. Expanding the entire residual gives

$$\begin{aligned} R^* T^{-1} R &= (C - TY_0)^* T^{-1} (C - TY_0) \\ &= C^* T^{-1} C - C^* Y_0 - Y_0^* C + Y_0^* T Y_0. \end{aligned}$$

Hence, with $K = C^* Y_0 + Y_0^* C - Y_0^* T Y_0$,

$$C^* T^{-1} C = K + R^* T^{-1} R \le K + s^{-1} R^* R. \tag{4}$$

There is no Galerkin equation in this identity. An arbitrary admissible trial works, and its quality affects only the residual size.

Let $D = (E - E_m)Y_0$. For every coefficient vector x the assumed action bound gives $\|Dx\| \le \eta \|Y_0 x\|$. Since the quadratic pairing is real by symmetry,

$$K - K_m = -\operatorname{Herm}(Y_0^* D) \le \eta Q_0, \qquad D^* D \le \eta^2 Q_0.$$

Also $R = r - D$. Applying scalar Young's inequality to the cross term in $\|rx - Dx\|^2$ for every x proves the matrix inequality

$$R^* R \le (1+\varepsilon) S_0 + (1+1/\varepsilon)\eta^2 Q_0.$$

Substitution in (4) proves $\nu C^* T^{-1} C \le \theta N$ from (3).

For completeness, the passage from this coefficient inequality to a form inequality is exact. Define the bounded map

$$L = \sqrt{\nu}\, T^{-1/2} C N^{-1/2}.$$

The coefficient statement is $L^* L \le \theta I$, equivalently $\|L\|^2 \le \theta$, and therefore $LL^* \le \theta I$ on the full Hilbert space. For an arbitrary $z \in \mathrm{Dom}(T^{1/2})$, test the latter inequality on $T^{1/2} z$. Since $T^{-1/2} T^{1/2} z = z$, this yields

$$\begin{aligned} \nu \langle C N^{-1} C^* z, z \rangle &= \|L^* T^{1/2} z\|^2 \\ &\le \theta \|T^{1/2} z\|^2. \end{aligned}$$

This is the claimed inequality on the entire form domain. It is an instance of standard majorization and factorization [2, Theorem 1], with its hypotheses supplied by the shift and $N > 0$. The physical matrix $C^* C$ never replaces N in this step.

Finally, if $K_m \le K_c + k_e I$, $Q_0 \le Q_c + q_e I$ and $S_0 \le S_c + s_e I$, all their multipliers in (3) are positive. Substitution therefore adds exactly the stated scalar error times I. The center matrices need not themselves be positive for this upper-bound argument. □

Taking separate largest eigenvalues of K_m and S_0 before adding them loses their correlation. Formula (3) retains that correlation until the final finite comparison. The difference is decisive in the actual calculation: the separate-norm certificate fails at $\mu = 1/20$, while the full matrix certificate passes there.

2.1 From a contraction to an energy–loss bound

Suppose on a closed quotient the original form satisfies

$$\mathfrak{a}(z) \geq E(z) - \langle Rz, z\rangle - \rho\|z\|^2, \qquad R = \nu C N^{-1} C^*.$$

If the preceding certificate gives $R \leq \theta(E + sI)$, with $0 \leq \theta < 1$, substitution retains the positive energy:

$$\begin{aligned} \mathfrak{a}(z) &\geq E(z) - \theta\Big(E(z) + s\|z\|^2\Big) - \rho\|z\|^2 \\ &= (1-\theta)E(z) - (\rho + \theta s)\|z\|^2. \end{aligned} \tag{5}$$

The parameter s is a shift of the positive operator. It is not a positive gap for the signed operator. Writing $s = \mu - \rho$ is a convenient parametrization with $\mu > \rho$; it gives the loss $\rho + \theta(\mu - \rho)$. A successful comparison improves the scalar floor relative to $-\mu$, while keeping $1 - \theta$ of E. If instead $\theta \geq 1$, the same algebra is valid but its energy coefficient is nonpositive. This explains the positive-retention restriction in the least-loss statement.

The construction has two logically separate tasks. First, an analytic and interval calculation supplies the finite-rank price R and the whole remainder ρ. Second, a residual comparison absorbs most of R into a shifted positive energy. A small finite signed eigenvalue alone performs neither task: it does not control the physical coupling to the rest of the Hilbert space. Both tasks are carried out below.

3 The actual theta–arithmetic certificate

Put $r_8 = (-d_8)_+$ and $R_8 = M_w r_8(D) M_w + 2p \otimes p$. The literal common signed form is

$$\mathfrak{a}(u) = E_8(u) - \langle R_8 u, u\rangle - \langle (J - J_8)u, u\rangle. \tag{6}$$

Since $|d_{8,+} - b| \leq |c_\Gamma| + \|k_8\|_\infty$, $E_8 - G$ is bounded, $\mathrm{Dom}(E_8) = \mathrm{Dom}(G)$ and $\mathrm{Dom}(E_8^{1/2}) = \mathcal{V}$. Let $S = I - q \otimes q/Q$ and E_Y be its positive form compression to $Y = q^\perp$. The projection preserves the graph domain.

The exact finite data consist of 2905 nonzero frequency cells $I_i = [k_i/64, (k_i + 1)/64]$, their midpoints $t_i = (2k_i + 1)/128$, and the exact upward binary64 mass bounds μ_i^+ for $\int_{I_i} r_8(t)\, dt/\pi$. Define

$$\begin{gathered} Be_i = \sqrt{\mu_i^+}\, S[w \sin(t_i x)], \qquad Be_p = \sqrt{2} S p, \\ C = BZ, \qquad N = Z^* Z, \qquad R_{56} = \frac{11}{10} C N^{-1} C^*. \end{gathered}$$

Every entry of the supplied 2906×56 matrix Z is interpreted as an exact dyadic rational. Its origin in a numerical solve is not an orthogonality or rank premise.

Proposition 3.1 (Verified realization data)**.** *The supplied interval certificates and analytical estimates establish*

$$(1-10^{-8})I \le N \le (1+10^{-8})I, \qquad C^*C \ge \frac{21}{10^8}N, \quad C^*C < \frac{32}{125}I,$$
$$\mathfrak{a}(z) \ge E_8(z) - \langle R_{56}z, z\rangle - \rho\|z\|^2, \quad \rho = \frac{979236349}{2000000000000}, \quad z \in \mathcal{V} \cap Y.$$

The actual physical rank is 56*. With all* 2469 *positive zeros through* 3000*, both signs and multiplicities included, the positive sampling operator* S_{3000} *satisfies*

$$\|(A_W - S_{3000})|_{\operatorname{ran} C}\| < 1.4 \cdot 10^{-18}, \qquad \mathfrak{a}|_{\operatorname{ran} C} \ge \frac{38}{10^{16}}I.$$

For $R_m = BB^*$ *and* $E_m = S_{3000} + R_m$*, the whole action price on this trial range is*

$$\|(E_Y - E_m)|_{\operatorname{ran} C}\| < \eta, \qquad \eta = 0.013155646076930.$$

Proof. We give the complete bridge and the finite enclosures which make the data reviewable. Supplement S1–S7 and the certificate files identified in the manifest give the interval construction, including all infinite allowances; none is a sampled-matrix positivity assumption.

The full support certificate proves $r_8(t) = 0$ for $|t| \ge 10000$, its positive-half-line mass is below 14, and $\|J - J_8\| \le t_8 = 32237/10^{12}$. A full theta moment gives

$$\int \|S[w\sin(tx)] - S[w\sin(t_i x)]\|^2 r_8(t)\, dt/\pi \le \delta_0 = 21/1024000.$$

Young's inequality on every cell, with parameter 1/10, gives $SR_8S \le (11/10)BB^* + 11\delta_0 I$, including the identical pole. The positive coefficient upper Gram Γ retains every frequency difference, sum, score and pole correlation. The full dense residual certificate proves $\|\Gamma - Z\Lambda Z^*\| < 3/12500$, with $\Lambda > 0$ exact dyadic. Compression to $(\operatorname{ran} Z)^\perp$ therefore gives $\|B(I-P)B^*\| \le 3/12500$, where $P = ZN^{-1}Z^*$. Adding the full prime tail yields $\rho = t_8 + 11\delta_0 + (11/10)(3/12500)$, exactly as displayed.

The actual feature Gram is $B^*B = D_\mu(G_0 - b_0 b_0^*/Q)D_\mu$, with $G_0(i,j) = [\widehat{w^2}(t_i - t_j) - \widehat{w^2}(t_i + t_j)]/2$, $G_0(p,i) = 2\int_0^\infty w^2 \sinh(x/2)\sin(t_i x)\, dx$, $G_0(p,p) = \tau$, $b_{0,i} = -t_i \Xi(t_i)$, $b_{0,p} = -1/4$. New outward integrals use the actual $Q \approx 11.06766574714$ and $\tau \approx 0.005712908530$, with complete allowances $2^{-250} + 2^{-180}$. The whole compression error is below $3.2 \cdot 10^{-9}$. A binary spectral locator is corrected by its full orthogonality defect and residual, giving the rational Gram bounds above. Thus $\kappa_B \le \sqrt{10^8/21} < 2182.178903$.

Native FLINT 3.6.0 zero enclosures exhaust its multiplicity count $N(3000) = 2469$ [8]. Disjoint actual roots prove finite-range simplicity. With $V_0 e_\gamma = w\sin(\gamma x)$ and $A_0 = V_0^* C$, the complete paired moments are $L = 2A_0^* A_0$ and $O = 4A_0^*(V_0^* V_0)A_0$. All 2469^2 correlations are kept. A new order-20, spacing-1/8 Taylor grid through 128 uses 320-node outward quadrature, the entire lattice and physical tails. The complete bounds are $\|\Delta A_0\|_F \le 2\cdot 10^{-12}$, $\|\Delta(V_0^* V_0)\| \le 3.2 \cdot 10^{-11}$, $\|\Delta L\| \le 10^{-11}$ and $\|\Delta O\| \le 3.1 \cdot 10^{-10}$. Direct rectangular-factor certification proves $L \ge 10^{-15} I$. The maximal actual midpoint is $T = 313317/128$. The full

contour and zero-count tails at height $L_0 = 3000$ are

$$\begin{aligned} e &= 10^6 \kappa_B^2 e^{T/4} Z_{1/4}(L_0), \\ d &= 11000 \kappa_B e^{T/8} Z_{1/8}(L_0), \\ Z_a(L) &= e^{-aL}\left[2L\log L + \frac{2(\log L + 1)}{a} + \frac{4}{a^2 L}\right]. \end{aligned}$$

They give $e < 3 \cdot 10^{-43}$ and $d < 1.4 \cdot 10^{-18}$, including every omitted/off-line zero and multiplicity. This proves the finite signed gap without asserting any infinite-block inverse.

Finally introduce the true-mass midpoint map on the same feature input. Its distance from the continuum feature map is at most $\sqrt{\delta_0}$. Thus its price differs from SR_8S in norm by at most $2\sqrt{0.255985513\delta_0} + \delta_0$. The entire weighted mass-deficit certificate adds less than 0.00855266, equivalently 427633/50000000, with zero pole deficit. On Y, $E_Y = A_W + SR_8S + S(J - J_8)S$. Adding t_8 and the whole action tail d proves the stated η. The columns are in $\mathrm{Dom}(G)$ by complete theta smoothing and verified score projection. This proves the domain and action premises. □

3.1 The coupled finite comparison

For each shift $\mu > \rho$, put $s = \mu - \rho$ and choose the supplied exact binary trial $Y_0 = CX$. It needs no certified solve accuracy. Let $G_B = B^*B$, $G_z = V_0^*V_0$, $Q_b = V_0^*B$. The whole residual is

$$\begin{aligned} r &= BD_1 + V_0 D_2, \\ D_1 &= Z(I - sX) - G_B ZX, \qquad D_2 = -2A_0 X, \end{aligned}$$

with exact Gram

$$\begin{aligned} r^*r = {} & D_1^* G_B D_1 + D_1^* Q_b^* D_2 \\ & + D_2^* Q_b D_1 + D_2^* G_z D_2. \end{aligned}$$

All $2469 \cdot 2906$ mixed correlations appear. In particular the component outside $\operatorname{ran} C$ is never removed. The certificate prices every input, inner/outer product, exact shift conversion and coefficient assembly. If e_d bounds the physical coefficient-map error and r_e the center Gram error, its actual Gram upper price is $r_e + 2\sqrt{\lambda_{\max}(S_c) + r_e}\, e_d + e_d^2$. This price is included before Lemma 2.1 is used. All small matrix locators use exact-input 192-bit Arb residuals and their polar-unitary correction. Only locators come from floating point.

The complete upper prices are exact rationals:

μ	ε	matrix price p	θ
1/20	2/3	100157725018/125000000000	400631/500000
1/25	1/2	450253581783/500000000000	225127/250000

Each p includes the full finite error prices and an additional 10^{-6} reserve. The strict margins to $\theta(1 - 10^{-8})$ are $9592169/(5 \cdot 10^{13})$ and $20685723/(25 \cdot 10^{12})$. Thus Lemma 2.1 proves $R_{56} \le \theta(E_Y + s)$. Inserting this into Proposition 3.1 gives $\mathfrak{a} \ge (1-\theta)E_8 - [\rho + \theta(\mu - \rho)]\|\cdot\|^2$,

which proves the two original matrix-only bounds on $q^{\perp}$. The enriched improvements in Theorem 1.1 follow in the next section. For a larger verified K, $z \in \mathcal{V} \cap Y$ and the literal kernel identity gives $\mathfrak{a}(u) = \mathfrak{a}(z)$. This transfers the full theorem without recertifying a changed frame.

4 Admissible residual enrichment and its full action

The original frame-only action error cannot be applied to a new residual direction. We instead use the actual dyadic trial $Y = Bb + V_0 v$ with 56 input columns, selected from the residual outside $\operatorname{ran} C$. No inverse or rank of an enlarged trial Gram is assumed. For every fixed frequency a, $w^2 \sin(ax)$ is Schwartz; the complete theta series proves the same smoothing for the pole and score terms. The bound $b(t) \le 17 + 9\log(1+|t|)$ therefore places each column in $\operatorname{Dom}(G) \cap q^{\perp}$. The existing positive physical Gram justifies the projection off $\operatorname{ran} C$ used in selecting the residual. Exact binary selected coefficients themselves define the trial, so rounded numerical orthogonality is not a premise.

Put $\ell_8 = m_\Gamma - k_8$, $L_8 = M_w \ell_8(D) M_w$, and let $\mathcal{E}$ denote the positive compressed operator previously called E_Y. Its full domain identity is

$$\mathcal{E} = SL_8S + R_m - 2Sp \otimes Sp + \Delta_R, \qquad \|\Delta_R\| \le \epsilon_R,$$

where $\epsilon_R = 2\sqrt{0.2559855 13\delta_0} + \delta_0 + 0.00855266$. This follows from $d_{8,+} = \ell_8 + r_8$ and the already proved whole-space mass and midpoint prices. It makes no low-zero action assumption on the new columns.

The completed outward Fourier calculation supplies a whole map $\widehat{A}$ with

$$\|\mathcal{E}Y - \widehat{A} - \Delta_R Y\| \le g, \quad g < 1.421821 \cdot 10^{-5}, \qquad \|Y\| < 2.159290.$$

Here the finite lattice has spacing $1/20$, frequency height 3300, and physical radius 3. The theta strip envelope bounds the complete Fourier row integral by $1.001591 \cdot 10^9$; contour shifting prices the alias below $1.532843 \cdot 10^{-5}$ before $\|w\|$. Complete node/location/rounding price is below $1.797275 \cdot 10^{-8}$, omitted frequencies below 10^{-88}, and physical tails below 10^{-157}. The physical action Gram retains the full $\widehat{w^2}(t-s)$ Toeplitz convolution, both Fourier shifts and the score subtraction. Native outward complex DFT radii are included. Supplement S9 gives all analytic envelopes and product guards; S10 gives the whole-action calculation and its error ledger. These are operator-map bounds, not sampled form positivity.

4.1 The entire action error in the resolvent comparison

For $s = \mu - \rho > 0$, set

$$G_Y = Y^*Y, \quad CY = C^*Y, \quad YA = Y^*\widehat{A},$$

$$CA = C^*\widehat{A}, \qquad AG = \widehat{A}^*\widehat{A}, \qquad H = C^*C.$$

These are exact physical matrices. Their outward numerical enclosures will be inserted only after the operator calculation. Define

$$\begin{aligned}
\widehat{R}_s &= C - \widehat{A} - sY, \\
K_s &= CY + CY^* - sG_Y - (YA + YA^*)/2, \\
S_s &= \widehat{R}_s^* \widehat{R}_s \\
&= H - CA - CA^* - s(CY + CY^*) + AG \\
&\qquad + s(YA + YA^*) + s^2 G_Y.
\end{aligned}$$

The appearance of AG is essential: $\widehat{A}$ takes values in the full Hilbert space, so this product measures action outside the trial span as well as inside it. Replacing AG by a compressed action product would no longer equal the displayed residual Gram.

Write the exact action discrepancy as

$$\mathcal{E}Y = \widehat{A} + \Delta_R Y + D, \qquad \Delta_R^* = \Delta_R, \quad \|\Delta_R\| \le \epsilon_R, \quad \|D\| \le g.$$

The two errors have different structure. The first is a bounded selfadjoint operator applied to Y, so its quadratic price keeps G_Y. The second is a complete map error on the coefficient input, so its worst quadratic price is a scalar multiple of I. For each coefficient vector x,

$$\begin{aligned}
|\langle \Delta_R Y x, Yx\rangle| &\le \epsilon_R \|Yx\|^2, \\
|\Re\langle Dx, Yx\rangle| &\le g\|Y\|_{\mathrm{upper}} \|x\|^2.
\end{aligned}$$

Consequently the exact completed-square quadratic is bounded above by $K_s + \epsilon_R G_Y + g\|Y\|_{\mathrm{upper}} I$.

For any $\tau > 0$, vector Young's inequality also gives

$$(\Delta_R Y + D)^*(\Delta_R Y + D) \le (1+\tau)\epsilon_R^2 G_Y + (1 + 1/\tau) g^2 I.$$

The true residual is $\widehat{R}_s - (\Delta_R Y + D)$. Apply Young a second time, now with $\beta > 0$, to separate this entire residual from its action error. Combining with (4) gives

$$\begin{aligned}
C^*(\mathcal{E} + s)^{-1} C &\le K_s + a_Y G_Y + a_S S_s + a_I I, \qquad (7) \\
a_Y &= \epsilon_R + \frac{(1 + 1/\beta)(1+\tau)\epsilon_R^2}{s}, \\
a_S &= \frac{1+\beta}{s}, \\
a_I &= g\|Y\|_{\mathrm{upper}} + \frac{(1 + 1/\beta)(1 + 1/\tau) g^2}{s}.
\end{aligned}$$

The selected value $\tau = g/(\epsilon_R \|Y\|_{\mathrm{upper}})$ is strictly positive. No rank or inverse of G_Y occurs anywhere in this proof. The coefficient space of Y has the same dimension as that of C, which is all that is required by these products.

4.2 Inserting finite matrix uncertainties

For clarity, let the exact matrices above have numerical centers, Hermitian for G_Y, AG and H, carrying operator errors $e_G, e_{CY}, e_{YA}, e_{CA}, e_{AG}, e_H$. With exact rational s, the triangle inequality gives the input-error parts

$$e_K \le 2e_{CY} + se_G + e_{YA},$$
$$e_S \le e_H + 2e_{CA} + 2se_{CY} + e_{AG} + 2se_{YA} + s^2 e_G.$$

The implementation additionally encloses the products and sums that form those centers, including conversion of the shift. Such assembly errors are added to e_K, e_S; they are not absorbed into the map error g without proof. If U_c is the center of $K_s + a_Y G_Y + a_S S_s$, then a sufficient final upper comparison is

$$\frac{11}{10}\left[U_c + (e_K + a_Y e_G + a_S e_S + a_I)I\right] \le \theta N.$$

This follows because $a_Y, a_S \ge 0$. An exact-input residual and orthogonality correction encloses the largest eigenvalue of the center. Its certified upper endpoint, after all scalar allowances, is compared with $\theta(1 - 10^{-8})$, using Proposition 3.1. Thus a numerical spectral locator alone is never the final premise. The full input allowances and product guards are given in Supplement S10–S11.

4.3 The certified parameter family

Supplement S11 certifies the endpoints for 231 rational (μ, β) candidates from these same cached Grams with independent exact-input Arb residuals, polar corrections, full assembly prices and rational endpoints. For $\mu = 1/25, \beta = 3/2$, the certified θ is exactly 0.867415467588; for $\mu = 1/20, \beta = 1277/512$, it is exactly 0.787541154611. Their exact losses are below the upward decimals in Theorem 1.1. For $\mu = 401/12800, \beta = 1913/2048$, $\theta = 499940839479/500000000000$ gives the third exact bound. Each final price is strictly below $\theta(1 - 10^{-8})$. Factorization and the original rest price prove Theorem 1.1. The accompanying finite verifier checks the exact candidate endpoints; the saved primitive Grams determine each selected comparison.

The central upper matrix has the form $A + \beta s G_Y + M/s$, $M \ge 0$. It is operator convex in s, but its derivative can have either sign. Thus the adaptive rational-neighborhood search makes no monotone bisection or interval exclusion claim. Only candidates with $\theta < 1$ enter the stated least-loss comparison; other tested endpoints can have a negative energy coefficient. Best tested in this class is not proved optimal. At the least-loss candidate the central price is about 0.997674; the complete finite uncertainty is below 0.002208. The actual whole residual and bounded continuum error remain substantive prices.

5 An actual equality obstruction to the fixed positive price

Define the first verified pole-null jet $h = (\Phi''' - \Phi'/4)/w$, $a = \langle h, q\rangle/Q$, and $k = h - aq$. Both h and q belong to $\mathrm{Dom}(G) \cap \ker A_W$; their transforms are polynomial multiples of

Ξ, vanishing at every actual zero, with the full signed explicit formula and graph-domain extension. Moreover $\widehat{wk}(t) = it(t^2 + 1/4 + a)\Xi(t)$, so $k \neq 0$.

Theorem 5.1 (Fixed-price equality barrier)**.** *On the score-only quotient, the actual vector k satisfies*

$$\langle R_{56}k, k\rangle - E_Y(k) > \frac{7}{5000}\|k\|^2, \qquad \mathfrak{a}(k) = 0.$$

Therefore any comparison $R_{56} \leq \theta(E_Y + s)$ with $0 < \theta \leq 1$ requires $\theta s > 7/5000$. Its scalar price $\alpha = \rho + \theta s$ must exceed 0.0018896181745*. This is a barrier for this fixed positive comparison, not a negative direction of the actual Weil form or a barrier after removing k.*

Proof. Two new complete meromorphic Arb integrals enclose $\langle h, q\rangle \approx 73.93892065366$ and $\|h\|^2 \approx 54152.69071824$; their full theta and physical allowances are less than 2^{-130}. They give $a > 6$ and $0 < D = \|k\|^2 < 55000$. The degree-five theta-atom jet polynomial is $-32y^5 + 360y^4 - 1056y^3 + 810y^2 - 90y$. Its complete lattice comparison bounds $|wh|/\Phi < 20000e^{6x}$. The entire doubled norm and cross tails beyond $x = 2$ are below $10^{11}\int_{54}^{\infty} t^8 e^{-3t}\,dt < 2^{-140}$. On $[0, 2]$ the omitted lattice and quotient expansion cost less than 2^{-170}; both costs are added before projection.

Every midpoint price term is nonnegative. Keep only the 635 cells with $t_i \leq 20$, replace a by 6 in the squared sine pairing, and evaluate the exact upward dyadic masses and actual $\Xi(t_i)$ with complex Arb arithmetic. The resulting sum is greater than 1875. Hence $r = \langle R_m k, k\rangle/D > 3/88$; the omitted pole and all other midpoint cells only increase this lower bound. To compare this midpoint energy with the actual continuum energy, let F_0 be the continuum feature map and F_1 the midpoint map using the true cell masses, on a common L^2 coefficient space. Their positive products are respectively SR_8S and R_t, and $\|F_0 - F_1\| \leq \sqrt{\delta_0}$. Since $0 \leq R_t \leq R_m$,

$$\begin{aligned}\sqrt{\langle SR_8Sk, k\rangle} &= \|F_0^* k\| \\ &\leq \|F_1^* k\| + \sqrt{\delta_0}\|k\| \\ &\leq \sqrt{\langle R_m k, k\rangle} + \sqrt{\delta_0}\|k\|.\end{aligned}$$

This vectorwise bound uses no sign assertion about the difference $SR_8S - R_m$. It is stronger for the present purpose than replacing that difference by a whole-space absolute norm.

Put $\varepsilon_c = 3/12500$ and $T = S(J - J_8)S$. The exact identities and inequalities on the kernel vector are

$$\begin{aligned}E_Y(k) &= \langle SR_8Sk, k\rangle + \langle Tk, k\rangle, \\ \langle Tk, k\rangle &\leq t_8 D, \\ \langle R_{56}k, k\rangle &\geq \frac{11}{10}\langle R_m k, k\rangle - \frac{11}{10}\varepsilon_c D.\end{aligned}$$

The first follows from the signed form and $\mathfrak{a}(k) = 0$; the last is the complete dense capture. Dividing by $D > 0$ and using the feature-map triangle bound proves

$$\frac{\langle R_{56}k, k\rangle - E_Y(k)}{D} \geq f(r), \tag{8}$$

$$f(r) = \frac{r}{10} - 2\sqrt{\delta_0 r} - \delta_0 - \frac{11}{10}\varepsilon_c - t_8.$$

For $r \geq 3/100$, its derivative $1/10 - \sqrt{\delta_0/r}$ is positive. The outward scalar comparison $f(3/88) > 7/5000$, together with $r > 3/88$, gives the first assertion. Every signed arithmetic term is present in the kernel identity; only nonnegative midpoint terms were omitted in obtaining the lower sum.

Finally suppose $R_{56} \leq \theta(E_Y + s)$ with $0 < \theta \leq 1$. Evaluation on this same vector, subtraction of $E_Y(k)$, and division by D give

$$\frac{7}{5000} < \theta s - (1-\theta)\frac{E_Y(k)}{D} \leq \theta s.$$

The nonnegativity of E_Y justifies the last inequality. Adding the fixed ρ proves the strict scalar-loss barrier in the theorem. Supplement S8 gives the scalar integral enclosures and complete tails. □

The barrier identifies lost equality cancellation caused by the positive midpoint enlargement. It redirects improvement toward the signed rest or a kernel-aware price, rather than unlimited decimal tuning. For $\Pi = I - k \otimes k/D$, set $B' = \Pi B$, $C' = \Pi C$. The exact update is

$$\begin{aligned} B'^* B' &= B^* B - b_k b_k^*/D, \\ C'^* C' &= C^* C - c_k c_k^*/D, \qquad b_k = B^* k, \quad c_k = Z^* b_k. \end{aligned}$$

Actual-zero orthogonality gives $V_0^* k = 0$; hence the sampling, zero Gram and zero/feature cross-Gram remain unchanged. The projected rank is still 56, since the existing sampling factor is injective. All columns retain their graph domain. The full projected action error costs $d\|CX\| + \eta_c\|C'X\|$; replacing the first norm by the second without proof would be invalid. Supplement S8 proves these kernel-aware identities. Its first complete numerical test has upper price greater than one and proves no improved sign.

6 Signed continuation and the remaining complement

The positive comparison above is sufficient for Theorem 1.1, but its equality barrier suggests retaining the signed cancellation. On the score quotient the exact operator-domain identity is

$$A_Y = SL_8S - 2Sp \otimes Sp - S(J - J_8)S.$$

Thus the continuum midpoint error cancels before approximation. For the same admissible enriched trial, Supplement S12 proves a complete signed action error below $11/10^6$, including the full prime tail and physical output. Supplement S13 gives a different cached signed approximant with error below $7/12500$, an actual positive five-dimensional subspace with form lower bound $\|f\|^2/5000$, and its full outside coupling. These secondary conclusions do not establish the sign of the complement. Supplement S14 states an inverse-free, fully coupled Schur criterion and identifies its unverified mixed-action hypotheses. No improvement beyond Theorem 1.1 is inferred from them.

7 Reproducibility, literature and what remains

The accompanying release includes exact binary trials and matrices, input and executed-source fingerprints, the analytical supplement and a bounded finite verifier. That verifier checks hashes, rational consequences, selected enriched comparisons reconstructed from cached matrices, and the robust-capture inverse/Rayleigh calculation. It accepts the upstream quadrature, zero-completeness, Taylor, large-Gram and whole-action error allowances as inputs; it does not regenerate those computations. The distinction between analytical proof, certificate inputs, finite replay and formal verification is recorded explicitly in the release verification report. Ordinary sampled positivity and an unpriced FFT are not certificate inputs.

The cancellation of a common bounded operator is elementary; its quantitative benefit here depends on the actual whole-action ledger. The factorization, coercive residual square and legal Schur inverse are standard [2, 3, 4]. Our specific result concerns the whole theta-weighted complex odd domain and the exact arithmetic realization. Zhu's compact-window theorem already certifies finite/tail coupling and odd-sector positivity [5, Theorems 1.1–1.2 and Section 6]. Specifically, Zhu's Theorem 1.1 lifts a finite Legendre block with explicit tail and coupling costs; Theorem 1.2 and Section 6 certify positivity for complex tests supported in $[-0.8, 0.8]$. The odd-sector floor there is $8.2065 \cdot 10^{-15}$. Here the test is wu on the whole line, the comparison norm is $\|u\|_{L^2(dx)}$, and the arithmetic tail is infinite. Compact support and theta multiplication impose different conditions, so neither numerical constant dominates the other theorem. Suzuki's Friedrichs realization on $L^2(-a, a)$ uses the same gamma multiplier, but its small-window positivity theorem has different support and norm hypotheses [6, Theorems 1.1 and 1.4]. Connes–Consani's archimedean trace lower bound has a finite loss under compact multiplicative support and a Fourier vanishing condition; it does not include the present full prime operator [7, Theorem 6.11]. These are structural precedents. The contribution claimed here is the explicit theta-weighted whole-domain certificate and the fixed-price equality obstruction. No first-method or exhaustive priority claim is made.

The zero-error task remains the original signed comparison $E_8(z) \geq \langle (R_8 + J - J_8)z, z\rangle$ on the required quotient. At fixed $\rho > 0$, even $s \downarrow 0$ in a successful sufficient comparison would retain the floor $-\rho$; Theorem 5.1 locates a stronger obstruction for this particular fixed price. The residual-enriched whole-action certificate is used in Theorem 1.1. The signed continuation certifies an actual positive five-dimensional subspace and its whole outside coupling; neither it nor the sharper signed-action estimate proves the required complement sign. The even sector and all-index Li transfer remain necessary. The published metric congruence retains its full $\beta\|M_\beta^{-1/2}v\|^2$ term and fixed-frame logarithmic conditioning [1, Proposition 7.1]; no $\beta \to 0$ control follows.

7.1 Formal verification scope

The accompanying executable Lean 4.32.0 project checks the central abstract argument, with mathlib pinned to a recorded commit. It proves the coercive inverse bound, exact full-residual identity and upper bound, the complete action-error comparison, and the coefficient-metric transfer on the entire form domain. The composed declaration `enriched_certified_retained_energy` passes from the full action decomposition and

its finite upper-price certificate to the energy–loss conclusion. Complex scalar restriction retains every complex vector and its Hermitian quadratic; it does not replace the coefficient metric by the physical Gram.

The kernel also checks the exact rational endpoint identities and pass predicates for all 231 saved records, and the three displayed numerical consequences under their explicit premises. The 22 audited declarations have only `propext`, `Classical.choice` and `Quot.sound` in their actual transitive axiom reports, with no `sorryAx`, custom axiom or native-evaluation extension.

This is conditional formal verification of the proof mechanism and exact arithmetic. Construction of the theta operators and closed forms, resolvent existence, admissibility of the actual trial columns, theta/Fourier identities, prime and zero tails, interval matrix enclosures, and original-form/kernel identification remain analytical inputs proved or certified in the written account. The entire theta–Weil specialization and the equality-barrier proof are not claimed as formalized. The included `lean/README.md` and `verification/FORMAL_CORRESPONDENCE.md` map the declarations to the published statements; the build and actual axiom output accompany the sources.

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